Compiled Solutions – Senior Division

1. **00-01 C3 Graph Theory**

The adjacency matrix is as follows 9

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}^2 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

2. **00-01 C3 Graph Theory**

The adjacency matrix is a follows: As shown at the left

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

3. **01-02 C3 Graph Theory**

The adjacency matrix is constructed as follows: 11

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

4. **01-02 C3 Graph Theory**

Number the vertices 1 through 10. Start with vertex 1 and draw the edges to vertices 2 through 10. There are 9. Next draw the edges from vertex 2 to vertices 3 through 10. There are 8. Repeating the process and summing the number of edges (9 + 8 + 7 + 6 + 5 + 4 + 3 + 2 + 1 = 45). 45

5. **02-03 C3 Graph Theory**

Label vertices from 1 to 9. Draw edges from 1 to 2, 1 to 3, … etc. There are 8 edges. Repeat starting at vertex 2. There are 7 edges. Repeat the process until the last edge from 8 to 9 is drawn. Sum 8+7+6+5+4+3+2+1. 36

6. **02-03 C3 Graph Theory**

A cycle is a simple path that begins and ends with the same vertex. 15
They are: BB, EDE, FCF, BDB, ABCDA, BCDB, AFEDA, AEDA,
ABDA, ABCFEDA, CFEDBC, ABCEDA, AFCDA, AFCEDA, BCEDB.

7. **03-04 C3 Graph Theory**

The cycles are: ABA, ABCDA, ACDA, ACBA, ADA, 9
BB, BCB, CDC and DD

8. **03-04 C3 Graph Theory**

$$\begin{bmatrix}1&1&0&1\\1&0&0&1\\0&1&1&1\\0&1&0&0\end{bmatrix}^2=\begin{bmatrix}2&2&0&2\\1&2&0&1\\1&2&1&2\\1&0&0&1\end{bmatrix}\bullet\begin{bmatrix}1&1&0&1\\1&0&0&1\\0&1&1&1\\0&1&0&0\end{bmatrix}=\begin{bmatrix}4&4&0&4\\3&2&0&3\\3&4&1&4\\1&2&0&1\end{bmatrix}$$ 36

There are a total of 36 paths of length 3.

9. **04-05 C3 Graph Theory**

The adjacency matrices are as follows after adding AC. 3

$$\begin{bmatrix}0&1&0\\0&0&1\\1&1&0\end{bmatrix}^2=\begin{bmatrix}0&0&1\\1&1&0\\0&1&1\end{bmatrix}\quad\text{versus}\quad\begin{bmatrix}0&1&1\\0&0&1\\1&1&0\end{bmatrix}^2=\begin{bmatrix}1&1&1\\1&1&0\\0&1&2\end{bmatrix}$$

This has 5 paths and the other has 8 paths of length 2.

10. **04-05 C3 Graph Theory**

The problem breaks down very quickly since there are no paths to C C
from A, B, or D. The starting town must be either C or E. Since the
starting town counts as the first town visited, the starting town must be C
to get a count of 5 ending at C.

11. **05-06 C3 Graph Theory**

There are a total of 10 paths of length 3. The math is as follows: 10

$$\begin{bmatrix}0&1&0&0\\0&0&0&1\\1&0&0&1\\1&1&0&0\end{bmatrix}^2=\begin{bmatrix}0&0&0&1\\1&1&0&0\\1&2&0&0\\0&1&0&1\end{bmatrix}\bullet\begin{bmatrix}0&1&0&0\\0&0&0&1\\1&0&0&1\\1&1&0&0\end{bmatrix}=\begin{bmatrix}1&1&0&0\\0&1&0&1\\0&1&0&2\\1&1&0&1\end{bmatrix}$$

12. **05-06 C3 Graph Theory**

The cycles from B are: BB, BCB, BCAB, BEAB, BEACB, BEADCB. 6

13. **06-07 C3 Graph Theory**

The cycles are: AA, ABDA, ACBDA, ACDA, ADA, BDB. 6

14. **06-07 C3 Graph Theory**

Cubing the adjacency matrix gives a totl of 36 paths of length 3. 36

$$\begin{bmatrix}1&1&1\\0&1&1\\1&1&0\end{bmatrix}^2=\begin{bmatrix}2&3&2\\1&2&1\\1&2&2\end{bmatrix}\bullet\begin{bmatrix}1&1&1\\0&1&1\\1&1&0\end{bmatrix}=\begin{bmatrix}4&7&5\\2&4&3\\3&5&3\end{bmatrix}$$

15. **07-08 C3 Graph Theory**

The cycles are: AA, ABA, ABCA, ACBA, ADCA, ADCBA, AEA, AEDCA, AEDCBA, ACA 10

16. **07-08 C3 Graph Theory**

$$\begin{bmatrix}0&1&1&0\\0&0&1&1\\1&0&1&1\\1&1&0&0\end{bmatrix}^2=\begin{bmatrix}1&0&2&2\\2&1&1&1\\2&2&2&1\\0&1&2&1\end{bmatrix}\bullet\begin{bmatrix}0&1&1&0\\0&0&1&1\\1&0&1&1\\1&1&0&0\end{bmatrix}=\begin{bmatrix}4&3&3&2\\2&3&4&2\\3&3&6&4\\3&1&3&3\end{bmatrix}$$

49

The sum of the elements in the solution is 49.

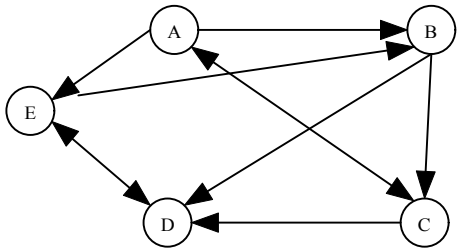
17. **08-09 C3 Graph Theory**

The cycles are: AA, ABA, ABCA, ABCDEA, ABCDA, ABCEA, ACA, ACDA, ACDEA, ACEA, ADA, ADEA, ADECA, AECA, AEA 15

08-09 C3 Graph Theory

This is the square of the adjacency matrix for the graph. 18
There are 18 paths of length 2.

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 1 & 1 & 3 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 2 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$



18. **09-10 C3 Graph Theory**

The original cycles are: AA, ABA, ABDA, ABDCA, ACA, ACBA, 3
ACBDA, ADA, ADCA, ADCBA. If edge DA were removed, then
ABDA, ACBDA, and ADA would not exist which is a total of 3.

19. **09-10 C3 Graph Theory**

The total number of paths of length 2 is 19. 19

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 2 & 2 & 1 & 2 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 3 & 1 \end{bmatrix}$$

20. **10-11 C3 Graph Theory**

These are the paths of length 3 and the most is 6 from A→B and A→C. A→B, A→C

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 3 & 2 & 2 \\ 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 1 & 1 \end{bmatrix} \bullet \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 6 & 5 \\ 3 & 5 & 5 & 4 \\ 2 & 4 & 3 & 3 \\ 2 & 4 & 4 & 3 \end{bmatrix}$$

21. **10-11 C3 Graph Theory**

The cycles are: B, BA, BC, BCD, BCDA, BCDEA, BD, BDA, BDEA, BEA 10

22. **11-12 C3 Graph Theory**

$$\begin{bmatrix}0&1&1&0\\0&0&1&0\\0&0&0&0\\1&0&1&0\end{bmatrix}^2=\begin{bmatrix}0&0&1&0\\0&0&0&0\\0&0&0&0\\0&1&1&0\end{bmatrix}\bullet\begin{bmatrix}0&0&1&0\\0&0&0&0\\0&0&0&0\\0&1&1&0\end{bmatrix}=\begin{bmatrix}0&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&0\end{bmatrix}$$

0 or NONE

23. **11-12 C3 Graph Theory**

There are 3 cycles from A: AA, AEA, and AEDA. 3

24. **12-13 C3 Graph Theory**

$$\begin{bmatrix}0&0&1&0&0\\1&0&1&1&0\\1&0&0&1&0\\0&1&1&0&1\\1&0&1&0&0\end{bmatrix}^2=\begin{bmatrix}1&0&0&1&0\\1&1&2&1&1\\0&1&2&0&1\\3&0&2&2&0\\1&0&1&1&0\end{bmatrix}^2=\begin{bmatrix}4&0&2&3&0\\6&3&9&5&3\\2&3&7&2&3\\9&2&8&7&2\\4&1&4&3&1\end{bmatrix}$$

A&B, A&E

Therefore no path of length 4 exists between A & B and A & E.

25. **12-13 C3 Graph Theory**

There are 5 cycles from C: CC, CABC, CAEC, CBC, and CEC. 5

26. **13-14 C3 Graph Theory**

$$\begin{bmatrix}1&1&1\\0&1&1\\1&0&0\end{bmatrix}^2=\begin{bmatrix}2&2&2\\1&1&1\\1&1&1\end{bmatrix}^2\bullet\begin{bmatrix}1&1&1\\0&1&1\\1&0&0\end{bmatrix}=\begin{bmatrix}4&4&4\\2&2&2\\2&2&2\end{bmatrix}$$

24

Cubing the adjacency matrix results in the number of paths of length 3.
Adding all the entries gives the total of 24.

27. **13-14 C3 Graph Theory**

There are 5 cycles from A: ABCA, ABDA, ABDCA, ADA, and ADCA. 5

28. **14-15 C3 Graph Theory**

These are the paths of length 3. There are none from A to C or C to A. A→C, C→A

$$\begin{bmatrix}0&1&1&0\\0&0&1&1\\0&0&0&1\\1&1&0&0\end{bmatrix}^2=\begin{bmatrix}0&0&1&2\\1&1&0&1\\1&1&0&0\\0&1&2&1\end{bmatrix}\bullet\begin{bmatrix}0&1&1&0\\0&0&1&1\\0&0&0&1\\1&1&0&0\end{bmatrix}=\begin{bmatrix}2&2&0&1\\1&2&2&1\\0&1&2&1\\1&1&1&3\end{bmatrix}$$

29. **14-15 C3 Graph Theory**

There are 7 cycles from B: 7
BAB, BADCB, BACB, BCB, BDEAB, BDEACB, AND BDCB.

30. **15-16 C3 Graph Theory**

$$\begin{bmatrix}0&1&1&0\\1&0&1&1\\1&0&0&1\\1&0&0&0\end{bmatrix}^2=\begin{bmatrix}2&0&1&2\\2&1&1&1\\1&1&1&0\\0&1&1&0\end{bmatrix}\bullet\begin{bmatrix}0&1&1&0\\1&0&1&1\\1&0&0&1\\1&0&0&0\end{bmatrix}=\begin{bmatrix}3&2&2&1\\3&2&3&2\\2&1&2&2\\2&0&1&2\end{bmatrix}$$ 30

Cubing the adjacency matrix results in the number of paths of length 3.
There are 30 of them.

31. **15-16 C3 Graph Theory**

There are 8 cycles from A: ABA, ACBA, ACDBA, ADCBA, ADBA, AEDCBA, AEDBA, AEA. 8